

Evaluation Statistics

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1 Setup

1.1 Import and Filter Data

Table 1: Entries excluded by timeout, minimum-duration, and dependency filtering.

phase	removal_reason	removed_entries
Phase 5	Elapsed time exceeded configured timeout plus grace period	5
R3	Recovery execution below 10 seconds; treated as not run	8
R4	Elapsed time exceeded configured timeout plus grace period	1
R5	Dependent recovery or stability entry removed with primary phase	3
R5	Recovery execution below 10 seconds; treated as not run	107
SR	Dependent recovery or stability entry removed with primary phase	3

1.2 Theme Settings

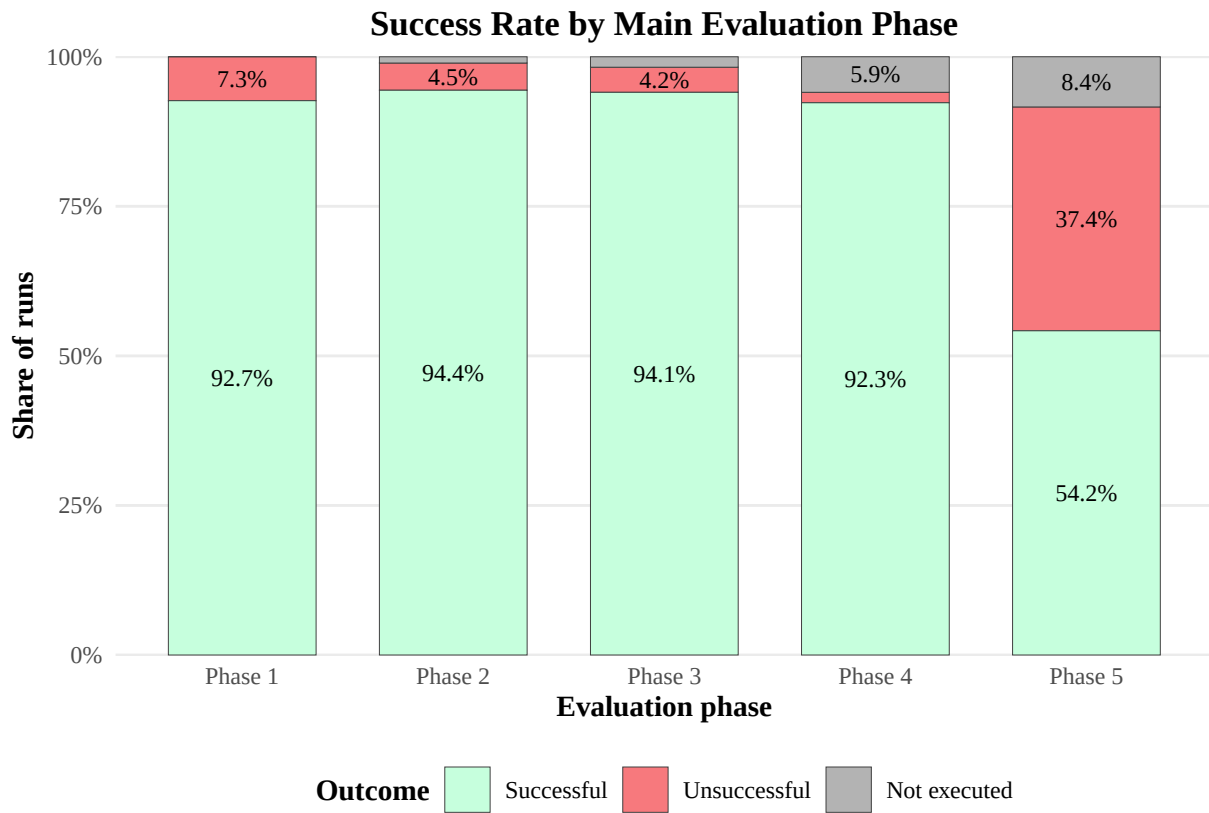
2 Success Rates

Table 2: Number of Valid Runs Recorded for Each Evaluation Phase

Phase	Number of runs
Phase 1	286
S1	125
R1	161
Phase 2.1	284
Phase 2.2	283
Phase 3	281
S3	262
R3	11
Phase 4	269
S4	161

Phase	Number of runs
R4	107
Phase 5	262
S5	127
R5	30
SR	278

2.1 Final Outcome by Main Phase



2.2 Direct, Recovered, Failed, and Not-Run Outcomes

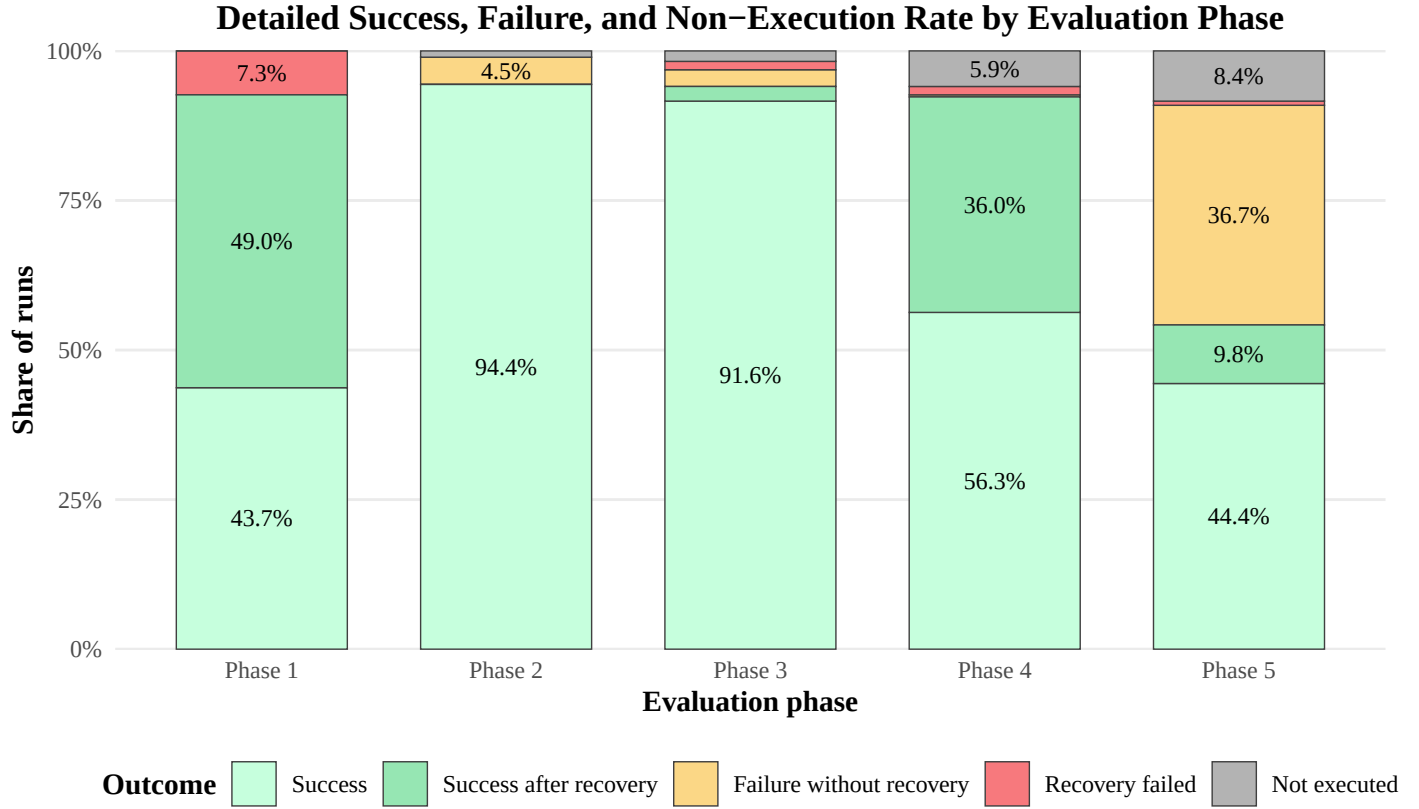
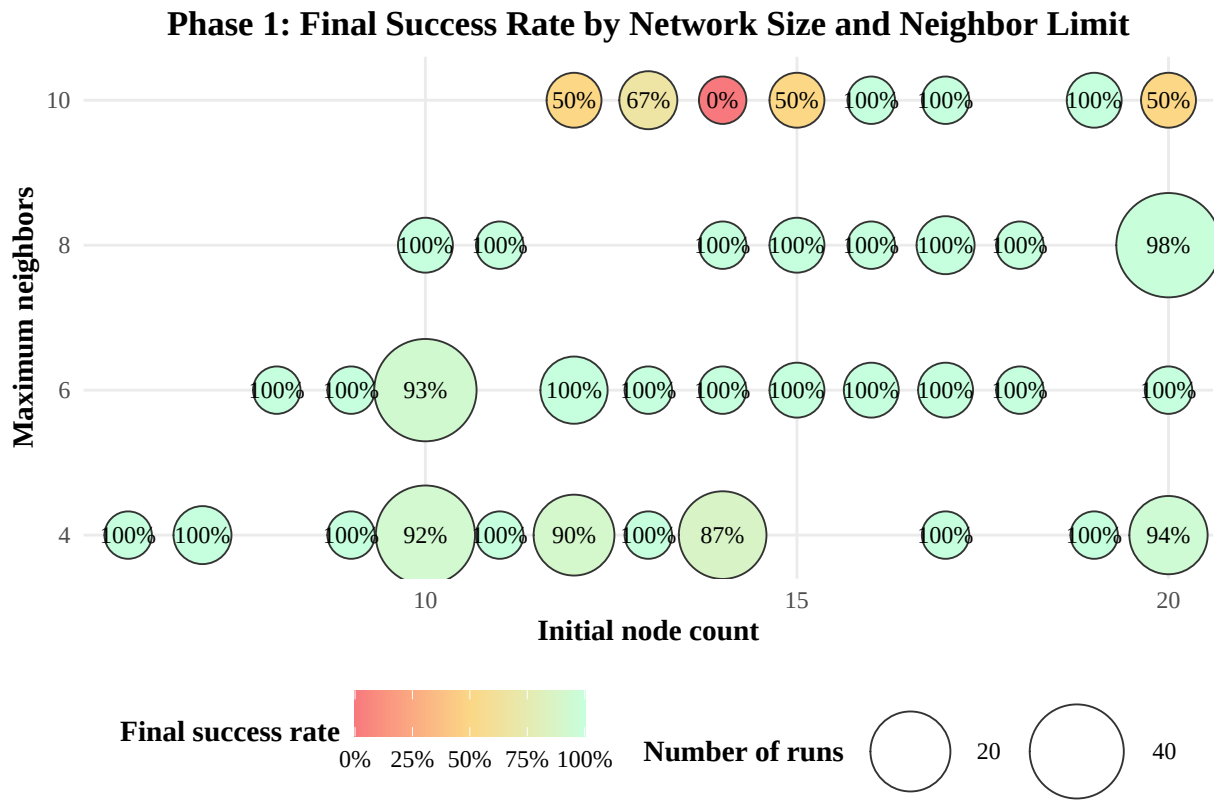


Table 3: Run counts by phase outcome, including phases that were not executed.

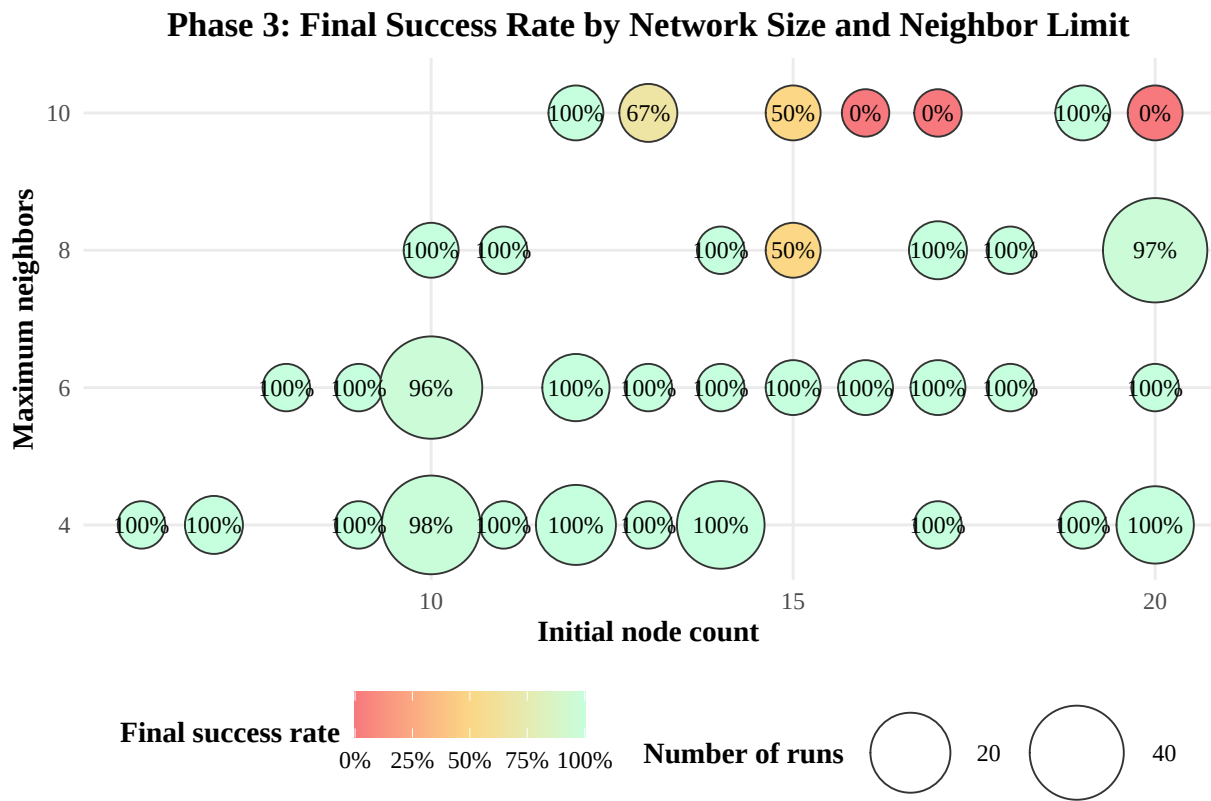
phase_group	Failure (no recovery)	Failure (with recovery)	Not run	Success (no recovery)	Success (with recovery)
Phase 1	0	21	0	125	140
Phase 2	13	0	3	270	0
Phase 3	8	4	5	262	7
Phase 4	1	4	17	161	103
Phase 5	105	2	24	127	28

3 Final Success Rate by Input Variables

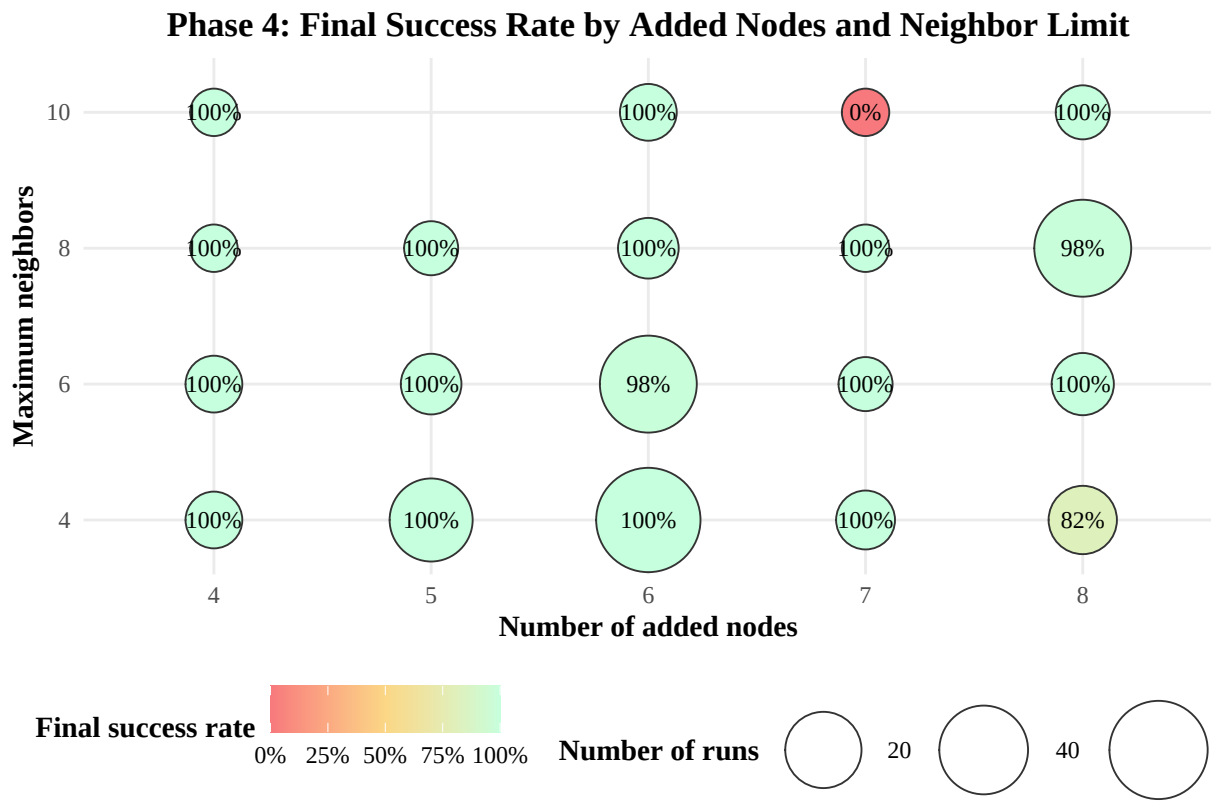
3.1 Phase 1: Initial Network Formation



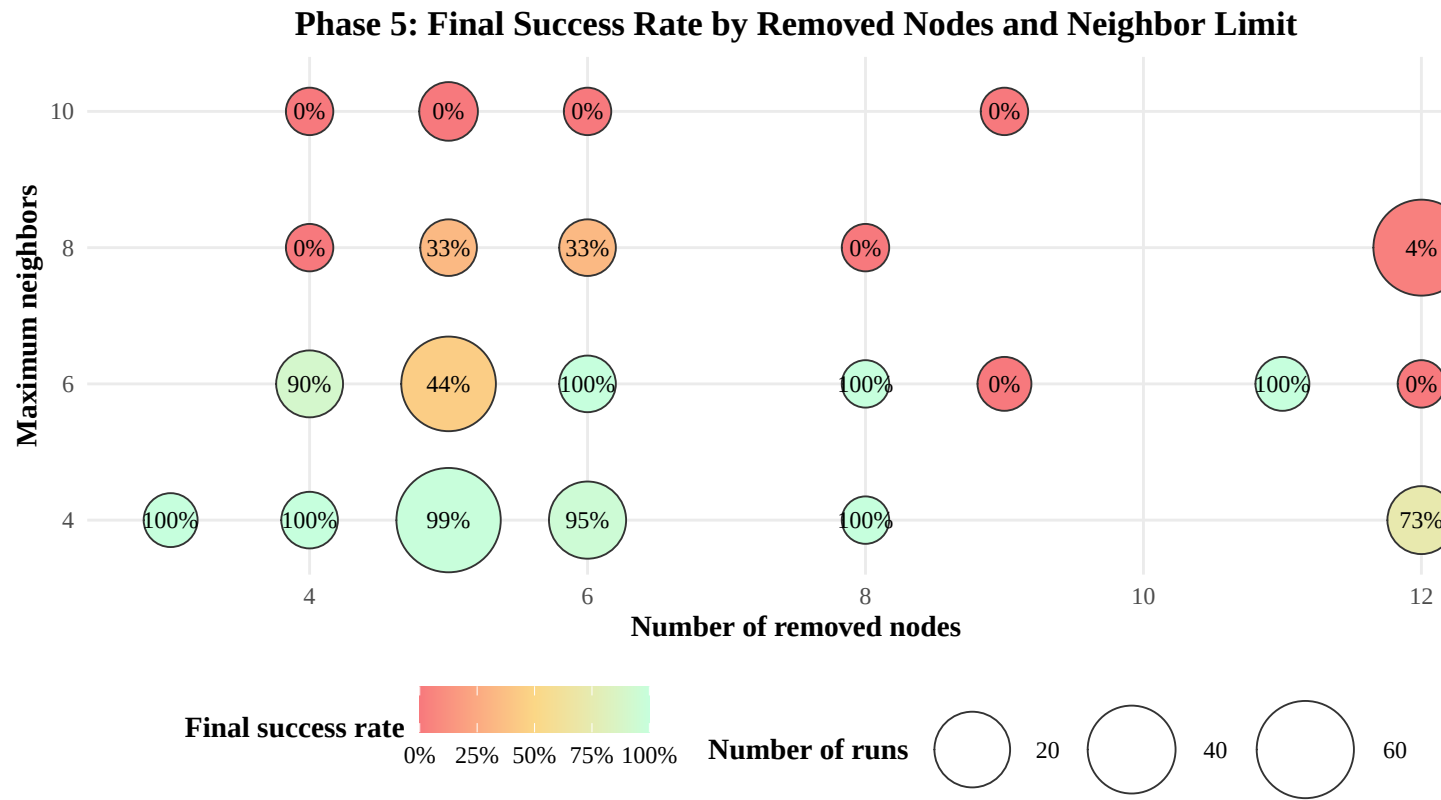
3.2 Phase 3: Single-Node Failure Repair



3.3 Phase 4: Adding Nodes



3.4 Phase 5: Removing Multiple Nodes

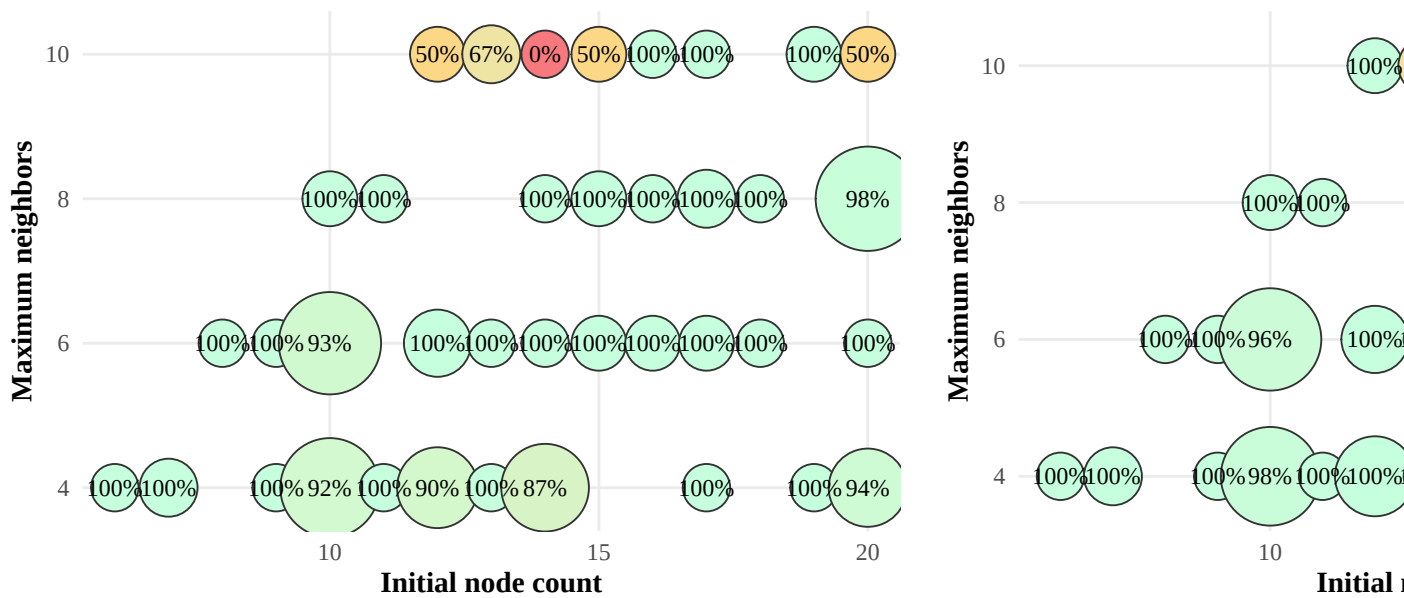


3.5 Combined Input-Variable Comparison

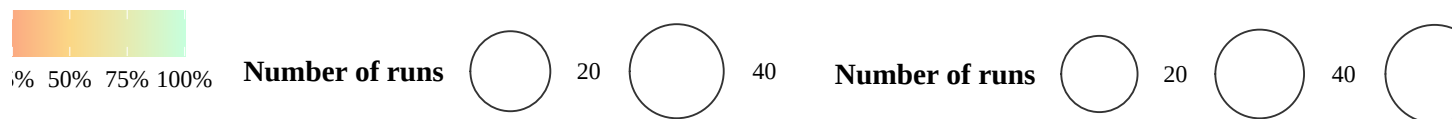
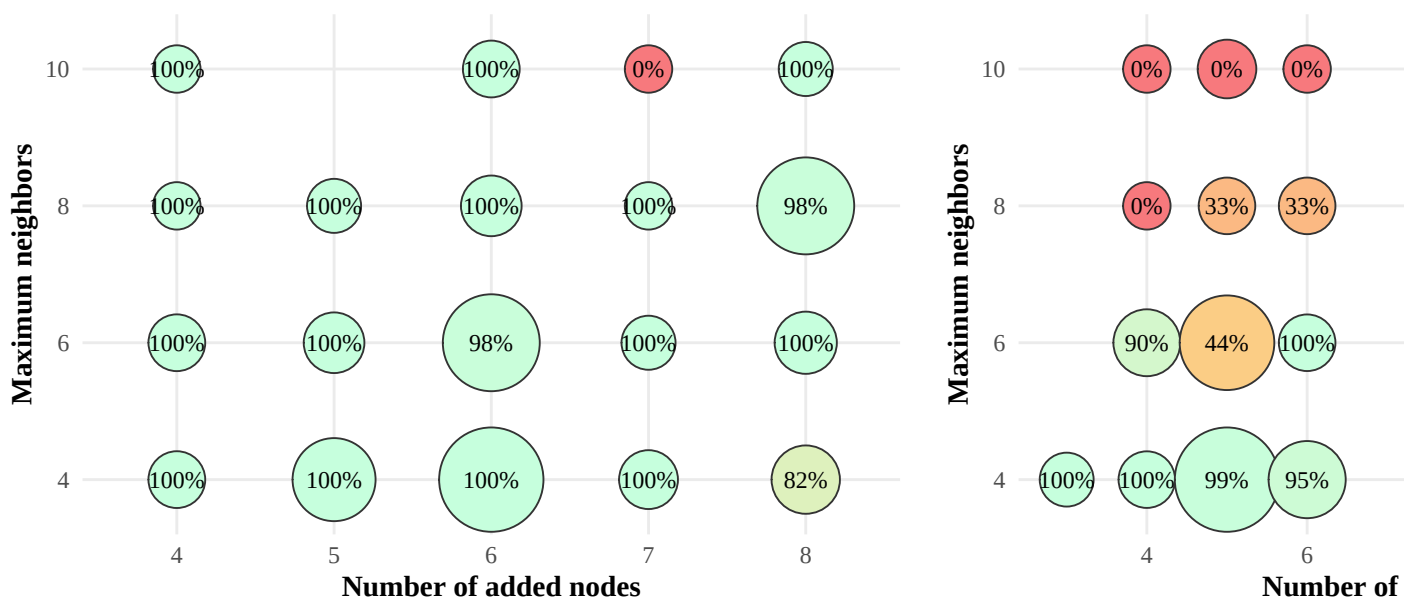
Final Success Rates Across Tested Network Sizes, Neighbor Limits, Node Addition

Final success includes successful replacement-based recovery; bubble size represents the number of re

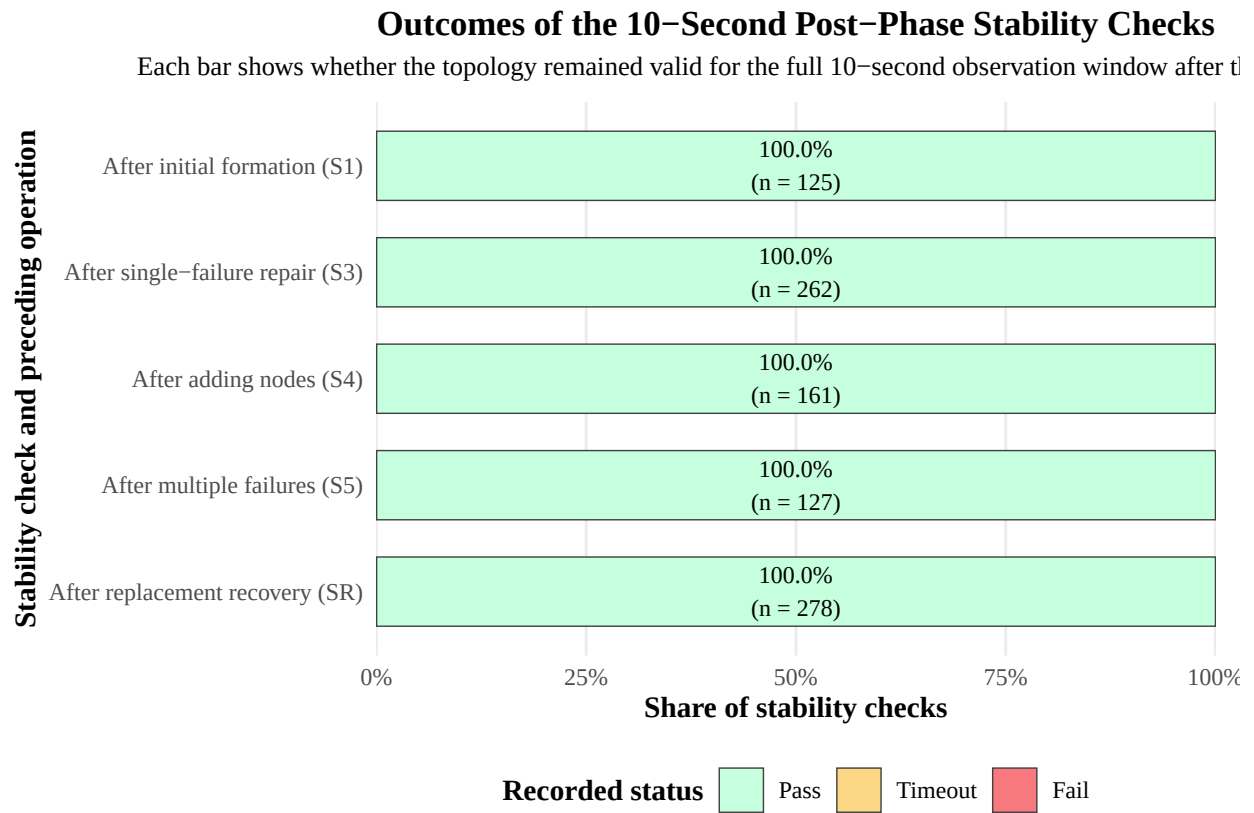
Phase 1: Final Success Rate by Network Size and Neighbor LimitPhase 3: Final Success Rate by



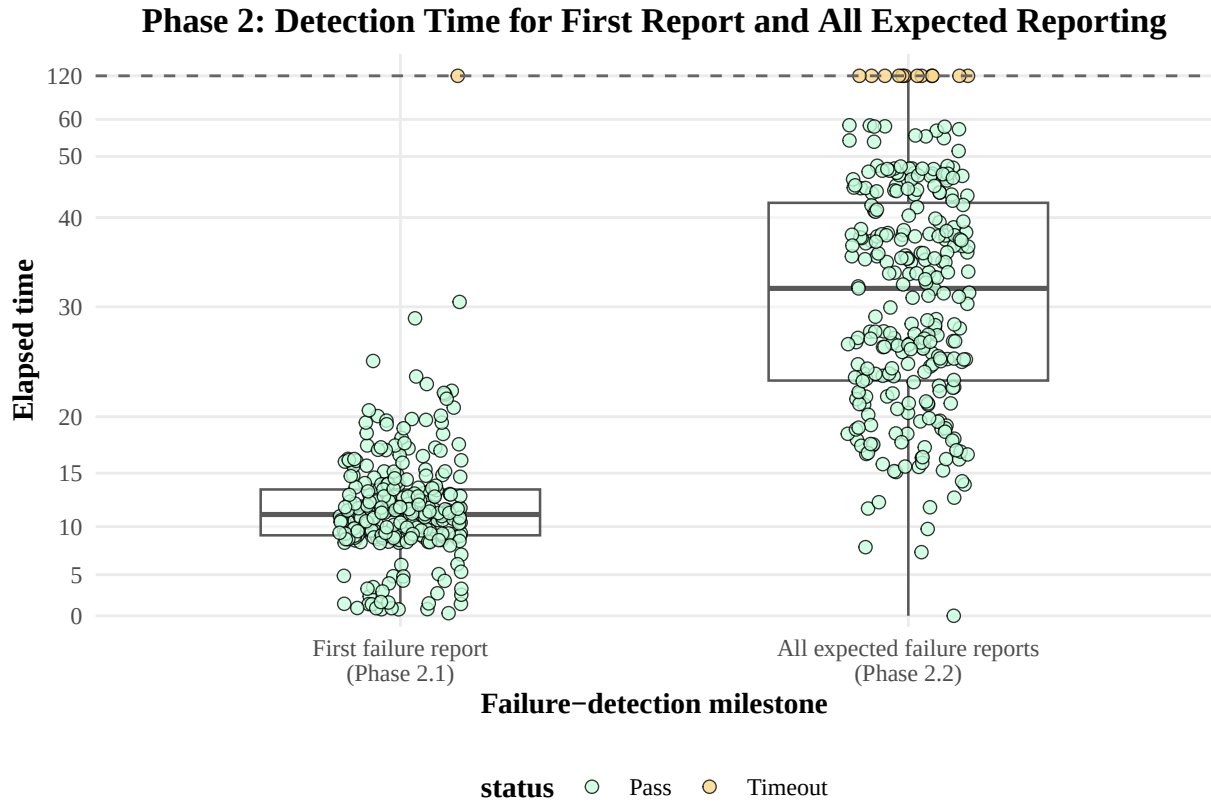
Phase 4: Final Success Rate by Added Nodes and Neighbor Link **Phase 5: Final Success Rate by Final Success Rate**



4 Stability Checks

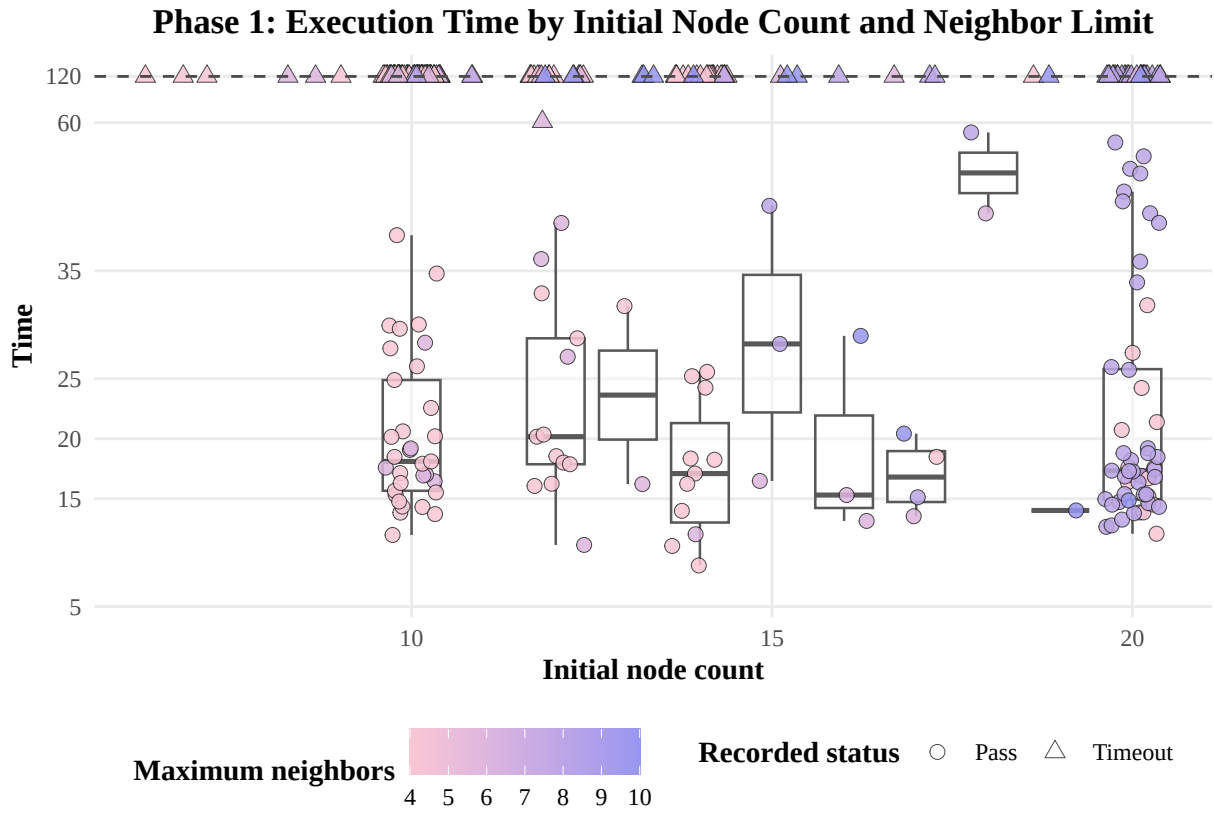


5 Failure Detection Time

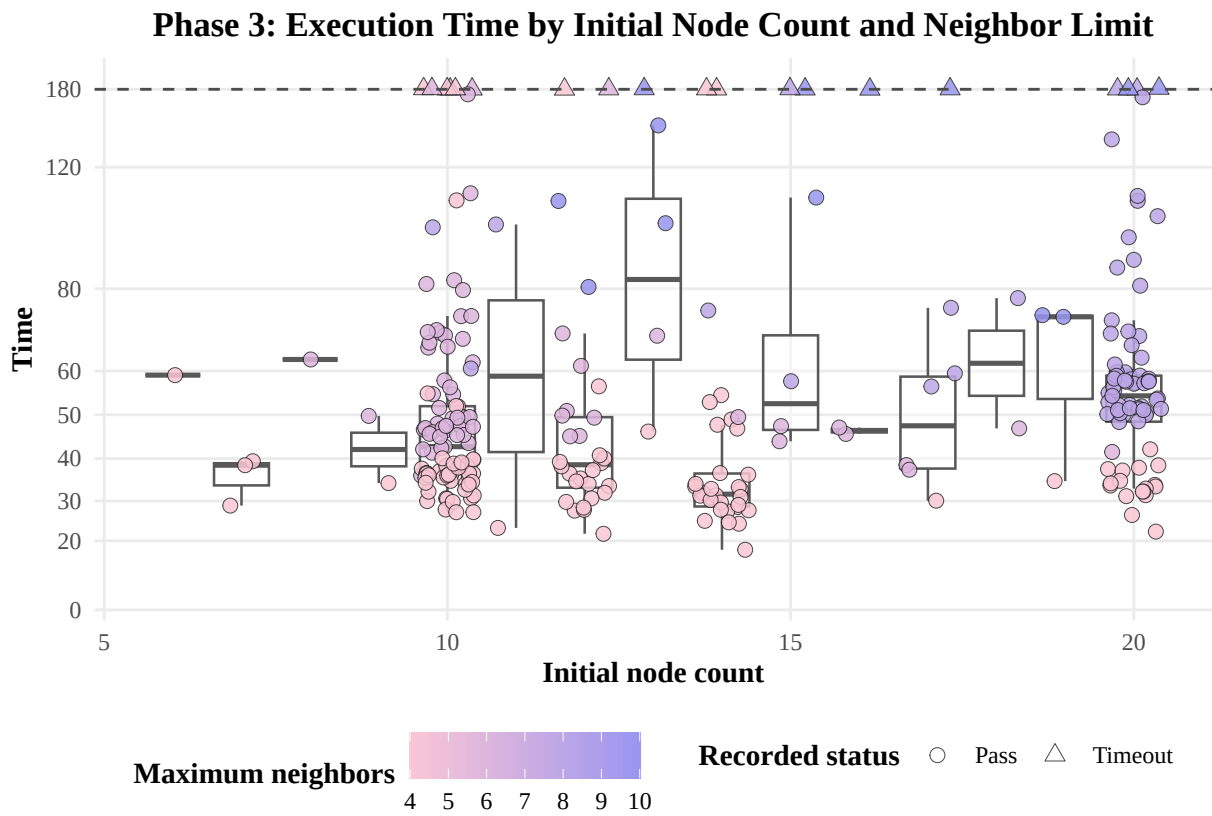


6 Main-Phase Completion Time and Input Effects

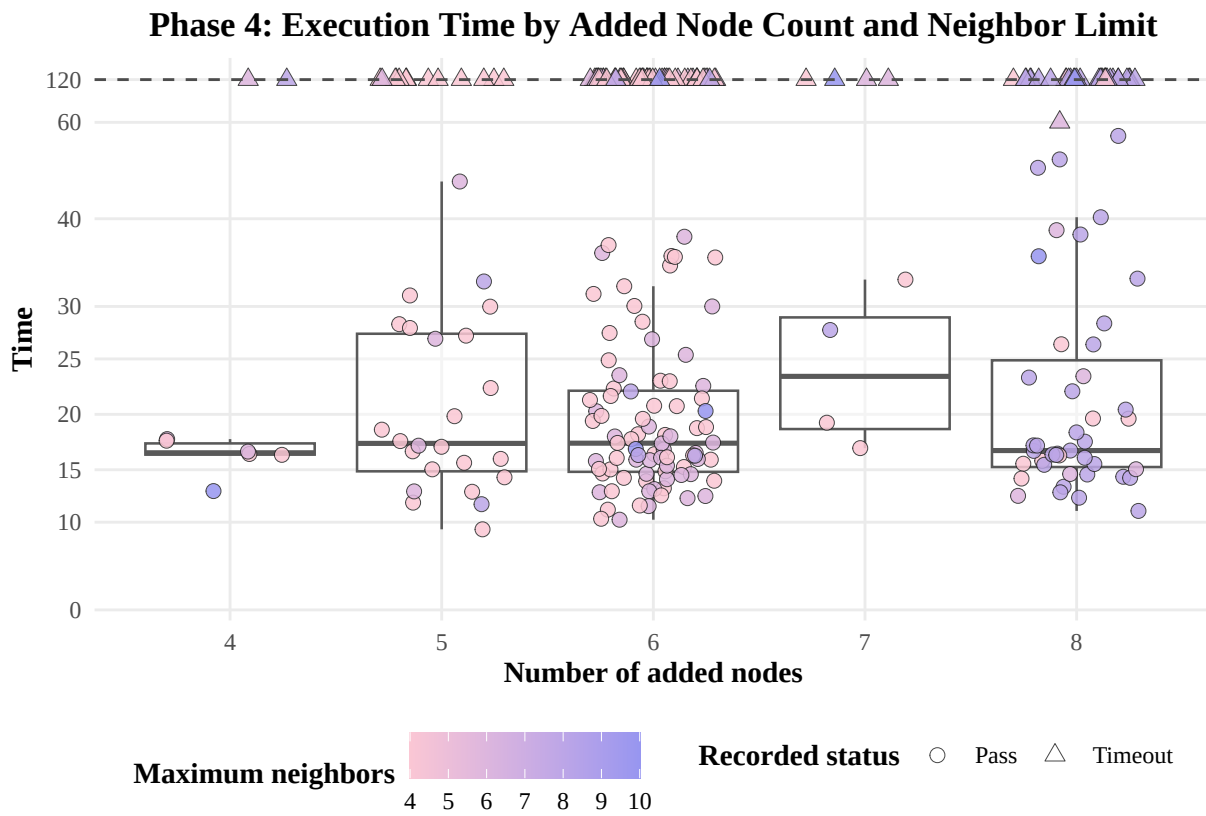
6.1 Phase 1: Initial Network Formation



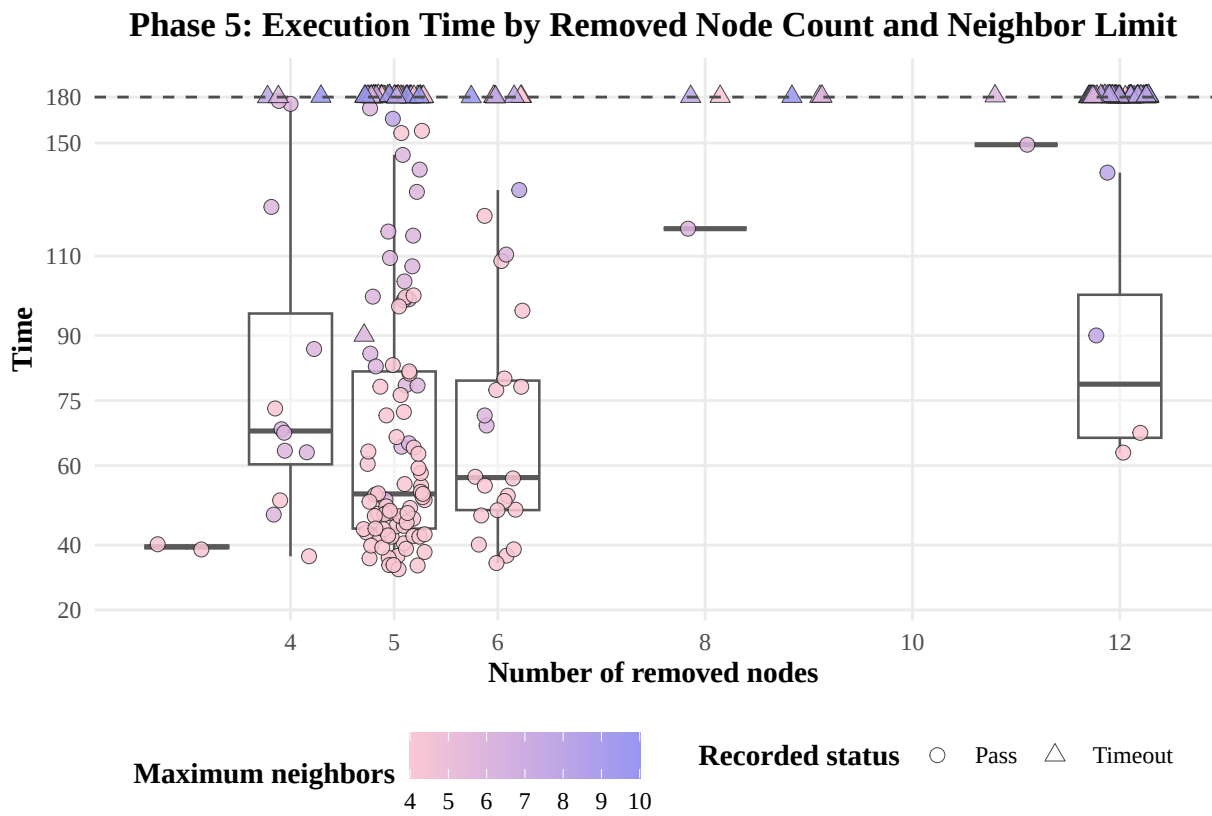
6.2 Phase 3: Repair After One Node Failure



6.3 Phase 4: Adding Nodes

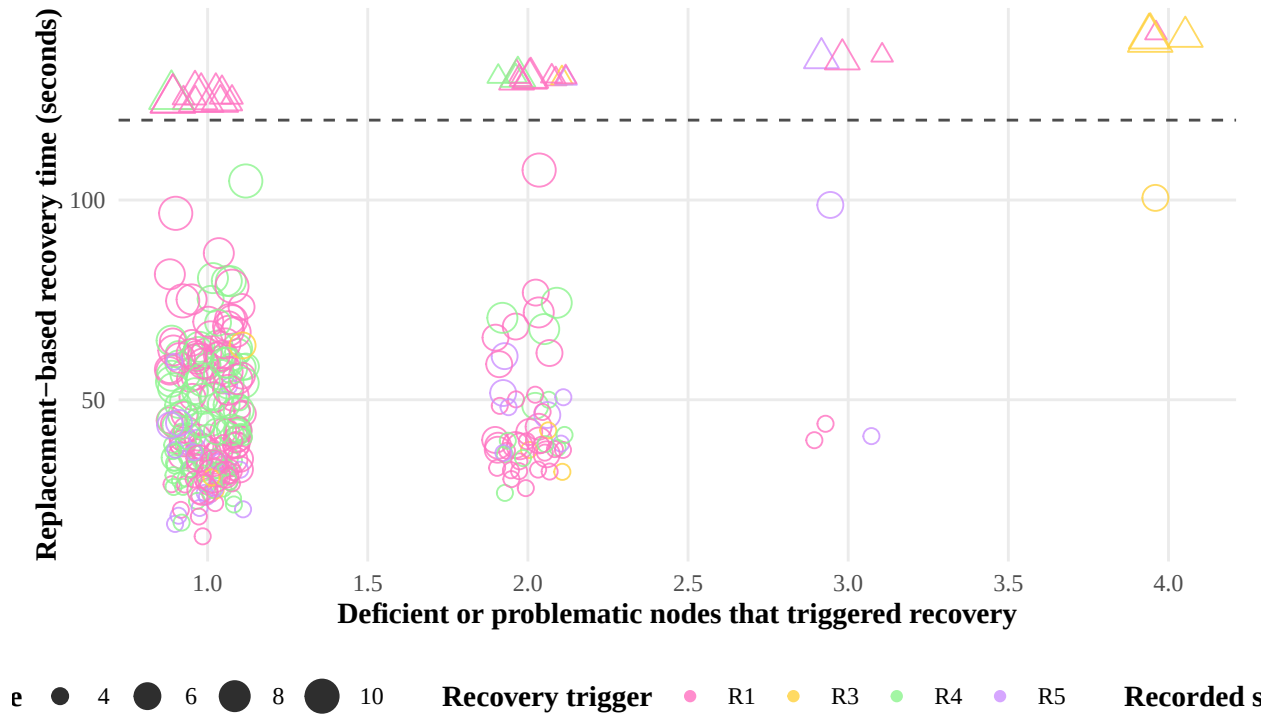


6.4 Phase 5: Repair After Multiple Node Failures



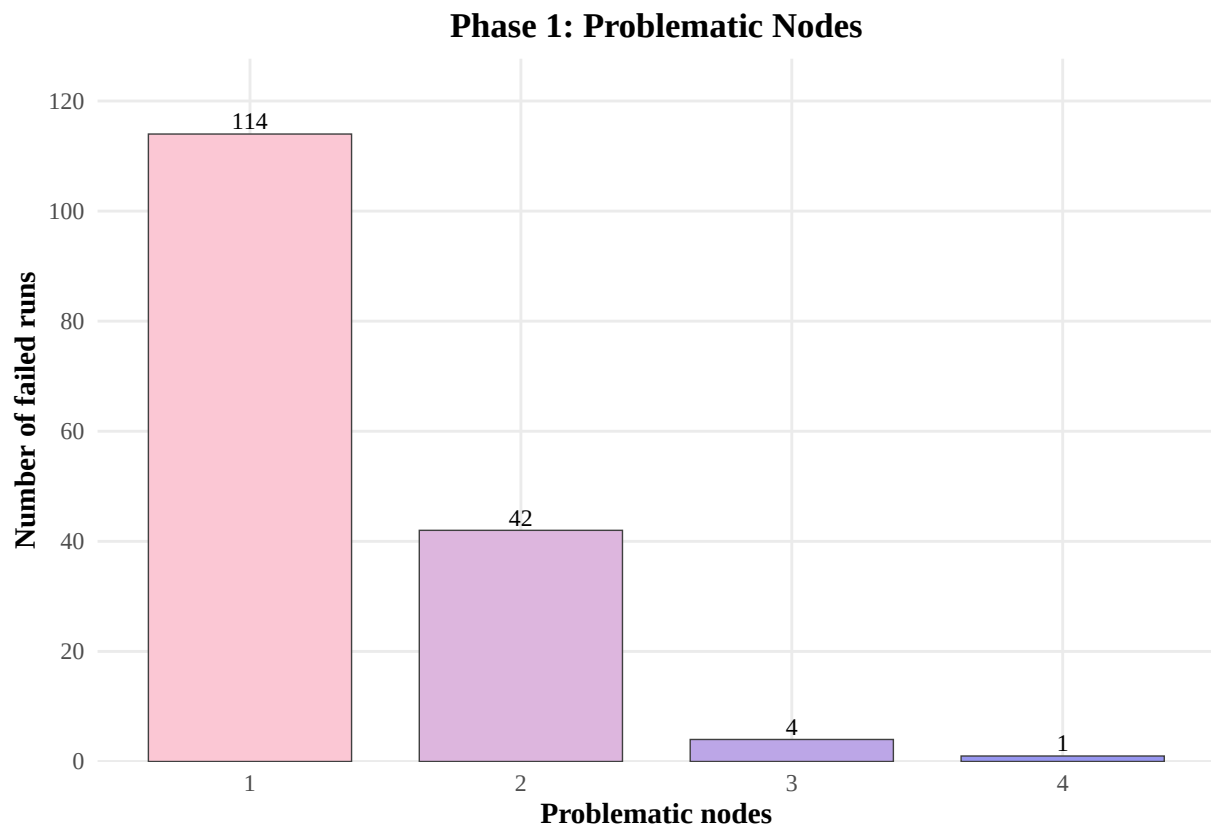
7 Replacement-Based Recovery Completion Time

Replacement-Based Recovery Time by Triggering Operation and Initial Problem C
 deficient/problematic nodes and wait for a valid topology. Dashed line: nominal 120-second timeout; dot-dashed line

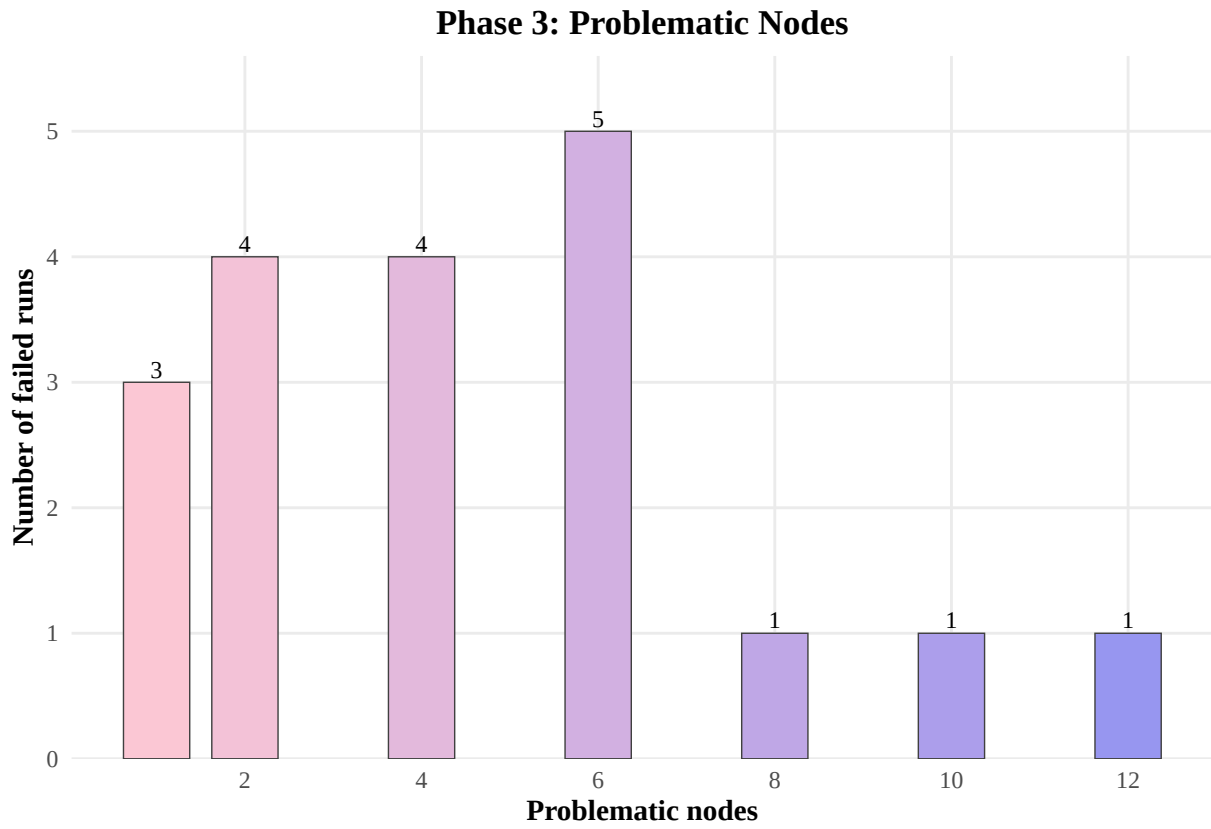


8 Problematic Nodes in Failed Main Phases

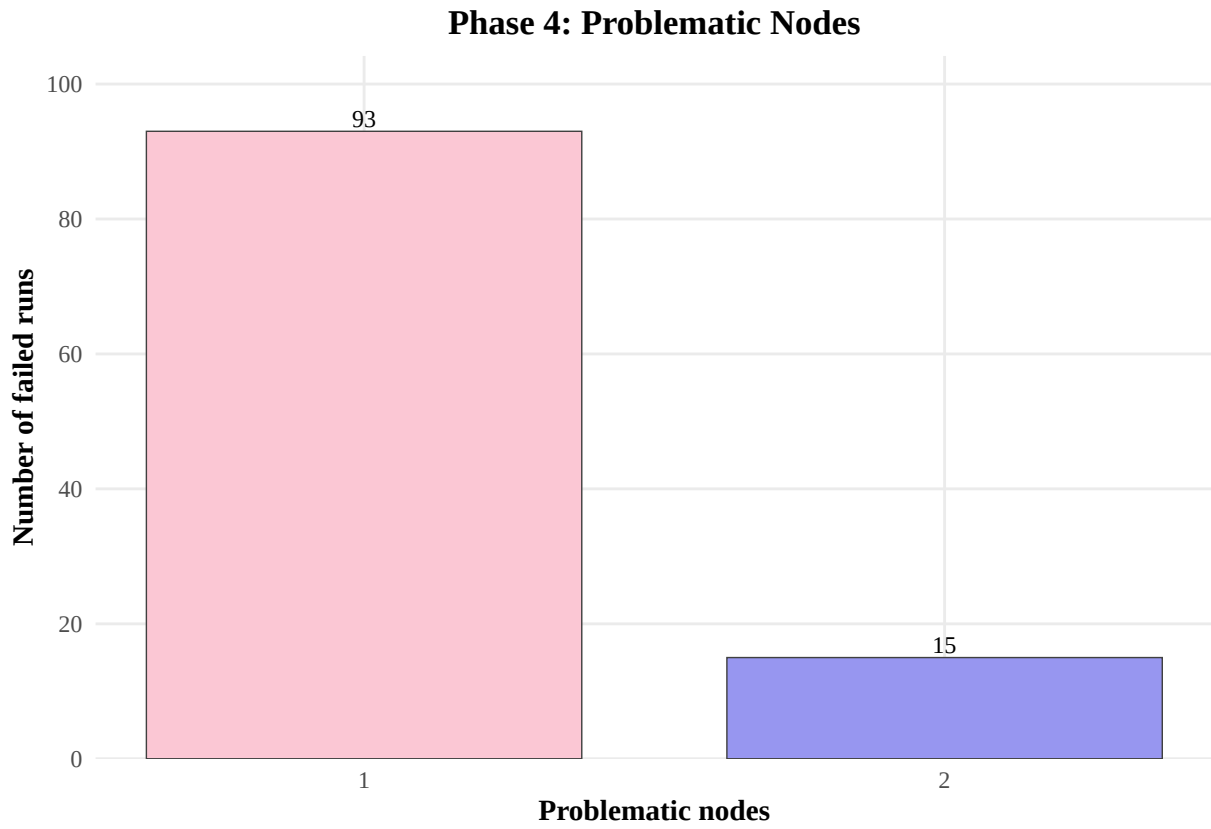
8.1 Phase 1: Initial Network Formation



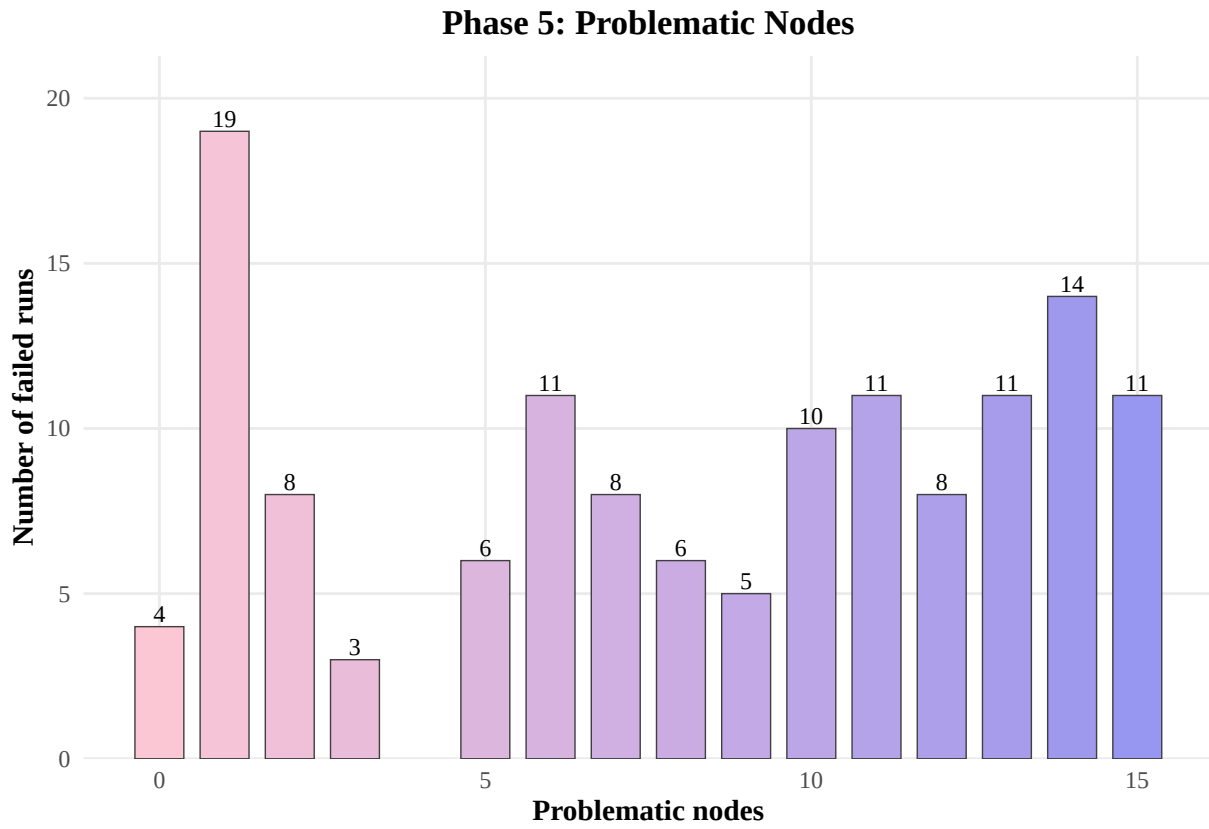
8.2 Phase 3: Repair After One Node Failure



8.3 Phase 4: Adding Nodes



8.4 Phase 5: Repair After Multiple Node Failures



9 Replacement-Based Recovery Success Rates

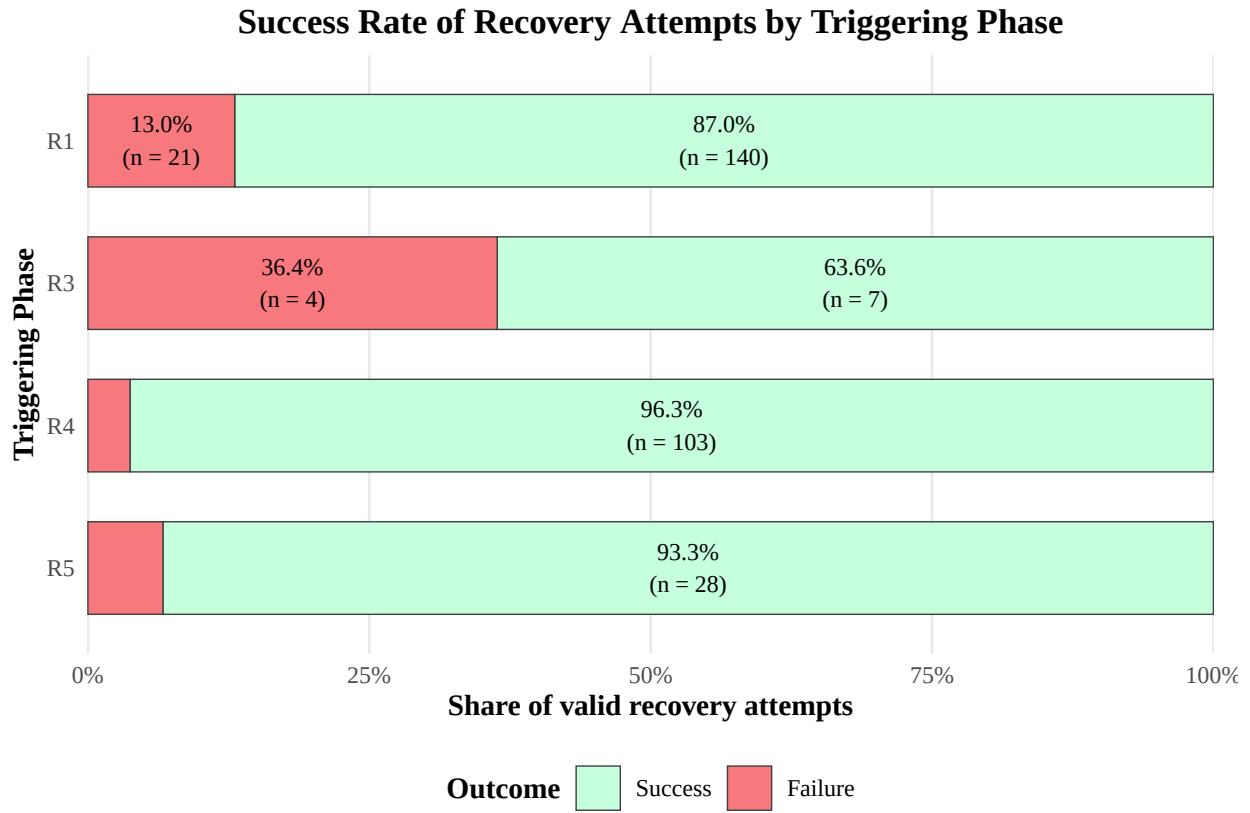


Table 4: Valid replacement-based recovery attempts after timeout filtering.

phase	attempts	passes	timeouts	failures	success_rate
R1	161	140	21	0	0.870
R3	11	7	4	0	0.636
R4	107	103	4	0	0.963
R5	30	28	2	0	0.933

Recovery Success Rates by Recovery Phase and Neighbor Limit

